

Rehabilitation of Achilles and patellar tendinopathies

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Achilles and patellar tendinopathies affect a broad range of the population and are difficult conditions to manage clinically. The pathology is persistent in the chronic tendon and can be considered to be failed healing. The exact cause of tendinopathy pain is unclear but may be related to changes in neurovascular structures.

Rehabilitation for Achilles and patellar tendinopathies is based on an exercise programme that aims to improve muscle–tendon function and normalise the pelvic/lower limb kinetic chain. This incorporates a programme for restoring and improving muscle strength, endurance and power and retraining sport-specific function.

Rehabilitation may take a prolonged period of time, both the athlete and clinician must be patient and persistent to maximise results from an exercise-based treatment.

Key words: Achilles tendinopathy; patellar tendinopathy; tendon healing; eccentric exercise.

INTRODUCTION

Achilles and patellar tendinopathies occur most commonly in people participating in sporting and physical activity^{1–4} but have also been reported in non-athletic populations.⁵ The exact aetiology of these conditions is unknown but is thought to consist of a combination of impact loading, genetic make-up, inefficient lower limb biomechanics and musculoskeletal function. Rehabilitation of Achilles and patellar tendinopathies can be difficult and prolonged and it requires both careful planning by the clinician and discipline from the patient to adhere to an often long rehabilitation programme.

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The pathology of chronic tendinopathy and the source of pain must be considered when planning a rehabilitation programme. The current knowledge of tendon pain, pathology and repair will be presented in this chapter together with an explanation of how this impacts on rehabilitation.

The essential components of the rehabilitation programme required to maximise success in managing these clinically difficult injuries will also be explained. Finally, there will be a discussion of the planning and implementation of a specific rehabilitation programme for Achilles and patellar tendinopathy, with particular emphasis on the indicators for the success or failure of rehabilitation.

TENDON PATHOLOGY AND REPAIR

Despite the anatomical proximity of muscle and tendon, the management of tendon pathology varies considerably from that of muscle injury. The differences in the management of muscle and tendon pathologies are reflected in their reparative responses to injury. Whilst the response to muscle injury follows a logical progression of inflammatory phase, muscle fibre regeneration and repair; tendon injury may not have an inflammatory stage and can result in a permanent state of pathology (failed healing).^{6,7} Histological evaluation of pathological tendons has demonstrated there is no evidence of prostaglandin-mediated inflammation⁸ although there may be some neurogenic inflammatory markers, such as neuropeptides, present.^{9,10}

In order to develop appropriate rehabilitation programmes for tendinopathies, it is important to understand tendon structure, pathology and repair. Normal tendons are well organised hierarchical structures that are predominantly made up of long strands of Type I collagen. The collagen is enveloped by ground substance that is mainly comprised of small proteoglycans with hydrophilic glycoaminoglycan chains, supplied by sparse, but adequate, neurovascular structures.⁶

Acute tendon injuries, such as lacerations, heal with a standard triphasic response of inflammation, proliferation and maturation, leading to functional scar formation. Overuse tendinopathy, however, does not follow the same pathway¹¹ and essentially results in long term disruption of the extracellular matrix.

Tendon pathology is characterised by four main changes in structure: (i) change in cell function, (ii) increase in ground substance, (iii) breakdown of collagen bundles and (iv) neurovascular proliferation (neovascularisation).¹² The activation and increase in the number of cells results in the increased production of essential extracellular materials such as ground substance and collagen. There is a change in the type of proteoglycans present in the ground substance in pathological tendons, with an increase in the larger proteoglycans such as aggrecan. The cells also produce Type III collagen, which is thinner and less capable of forming bundles than Type I collagen. The combination of inferior Type III collagen and the excessive amount of ground substance, leads to a disruption in the structure of the tendon and affects its ability to absorb forces.¹³

Tendinopathy is also associated with an increase in blood vessels and nerves within the tendon. Whilst this neovascularisation may appear to be part of the normal process of soft tissue repair, the presence of these neovessels and their associated nerves is thought to play an important role in tendinopathy-related pain. The presence of these vessels has been linked to symptomatic Achilles and patellar tendinopathies^{14–18}, as will be discussed later in this chapter.

These four components associated with tendon pathology are also part of the repair process, therefore, tendinopathy can be defined as a failed healing response.^{6,7}

This impaired healing results in a breakdown of the tendons' key function: their load absorbing and transmitting properties.¹¹

TENDON PAIN

The cause of pain in tendons is not known. The fact that tendon pathology is present in both symptomatic and asymptomatic individuals indicates that there may be specific aspects of histopathology that cause pain^{19,20} or that the source of pain is from structures independent of the pathology. Neovascularisation, a core part of tendon pathology, has provided a potential explanation into the pain mechanism associated with tendinopathy.

Blood vessels (imaged using Doppler ultrasound) are present in pathological Achilles and patellar tendons but not normal tendons.^{18,21} The presence of these vessels is considered to be associated with pain for three reasons. Firstly, neovascularisation that is evident on Doppler ultrasound is correlated with greater pain and poorer function.²² Secondly, the injection of sclerosing agents into the neurovascular bundles has produced good results so far, with improved function and a decrease in pain scores.¹⁴ Finally, patients with tendinopathies that have improved following treatment have also demonstrated a decrease in neovessels.²³ However, not all tendons with visible vessels are painful and, vice versa, neither has the presence of vessels been shown to affect long term outcome.^{24,25}

Whilst there is evidence that neovascularisation is an important component of painful Achilles and patellar tendinopathies, it is unclear how this occurs. Bjur et al (2005) have demonstrated that nerves were closely associated with the neovascularisation found in pathological tendons.²⁶ In addition, neurokinin-1-receptor, which is closely associated with the neuropeptide substance P, has been found in the vascular wall of the neovessels.²⁷ It is possible that the presence of neuropeptides may be indicative of 'neurogenic inflammation' within the tendon. Current research into pain fibres in normal and abnormal tendon may explain tendon pain in the near future.

AETIOLOGY OF TENDINOPATHY

The aetiology of Achilles and patellar tendinopathies is multifactorial; therefore it is important to establish if these factors are associated with the patient's tendon pain. Any identified aetiological factors need to be managed as part of a rehabilitation programme.

Overuse

During running and jumping activities, the Achilles and patellar tendons are subjected to forces ranging from six to 14 times bodyweight.^{28–30} Repetitive tendon loading during running and jumping activities is an important aetiological component of Achilles and patellar tendinopathies. Ferretti et al¹ found that athletes participating in more than three training sessions per week were more susceptible to patellar tendinopathy than those participating in less than three sessions per week. Similarly, a greater number of hours of sport per week have also been shown to be associated with patellar tendinopathy.³¹ These associations of training volume and frequency with the development of patellar tendinopathy may explain why elite athletes are more likely to develop patellar tendinopathy than recreational athletes.³²

Although repetitive tendon loading is an important factor in the development of tendon pathology, it is unclear whether it is entirely responsible for tendon pain.

A retrospective study of partially ruptured Achilles tendons demonstrated that physical activity was not necessarily associated with the degree of pathology.³³ Furthermore, Cook et al²⁰ reported that whilst athletes exposed to a moderate level of tendon loading demonstrated predictable tendinopathy changes, those changes were equally present in symptomatic and asymptomatic athletes.²⁰ This would suggest that there is a level of overuse or repetitive load that leads to tendon pathology but not symptoms.

Altered lower limb function

Lower limb tendinopathies have been shown to be associated with changes in lower limb function, although it is not known if these changes precede, or are a result of, the tendinopathy. Deficiencies in muscle–tendon function such as calf and quadriceps weakness in patients with Achilles tendinopathy and patellar tendinopathy, respectively, are commonly seen.^{31,34} These changes in muscle–tendon function can alter the coordination of movements of the hip, knee and ankle joints during functional weight-bearing tasks (kinetic chain function) and will impact on the patients' ability to develop power, absorb loading forces and will also alter running and jumping technique. Silbernagel et al³⁴ demonstrated that patients with Achilles tendinopathy have poorer hopping ability on the symptomatic leg, whilst Cook et al³⁵ found that patellar tendon abnormalities observed on ultrasound scanning were associated with a decreased vertical jump. Therefore improving muscle–tendon function and kinetic chain function should be of primary importance when developing a rehabilitation programme.

Biomechanical factors

Biomechanical factors that may lead to higher ground-reaction forces or altered movement patterns during single leg loading have also been linked to Achilles and patellar tendinopathy. For example, a decreased range of ankle dorsiflexion leads to greater ground-reaction forces³⁰ and increases the risk of developing Achilles³⁶ and patellar tendinopathies.³⁷ Similarly, decreased flexibility in the hamstring and quadriceps muscle groups has been found to be related to patellar tendinopathy.³⁵

Foot posture, particularly increased foot pronation, has been linked to Achilles tendinopathy³⁶, however there is little empirical evidence to support this due to the difficulty of reliably measuring foot biomechanics. Clinically, we believe that substantial malalignments of the foot need to be addressed as they have the potential to increase ground-reaction forces and tendon loading.

Intrinsic factors

Intrinsic factors determined by an individual's genes, such as gender, are factors that cannot be clinically altered, however it is useful for the clinician to appreciate their link to tendinopathy. There are several large cohort studies of patellar tendinopathy that have demonstrated a greater proportion of males compared to females with asymptomatic tendon pathology, suggesting that males are more likely to develop tendinopathy.^{20,38} Also, oestrogen has been shown to protect tendons from pathology.³⁹ There is also evidence that genetic factors may be related to tendinopathy. Mokone et al⁴⁰ found that there was a different distribution of the Tenascin-C gene (an extracellular matrix glycoprotein) and the Type V collagen gene in individuals with symptomatic Achilles tendons compared with those who were asymptomatic.

ASSESSMENT OF ACHILLES AND PATELLAR TENDINOPATHY

Achilles tendinopathy occurs most commonly at the midportion of the tendon and less frequently at the calcaneal insertion.^{2,41} Patellar tendinopathy, however, occurs most commonly as an enthesopathy at the attachment to the inferior pole of the patellar rather than the mid-tendon or distal insertion.⁴² A detailed assessment of an individual presenting with Achilles or patellar tendinopathy is essential as it dictates the content of the rehabilitation programme. Here, we will briefly explain some of the key aspects of the clinical assessment necessary to establish a diagnosis and plan the rehabilitation.

History

Both Achilles and patellar tendinopathies are characterised by a history of an insidious onset of pain, often associated with a change in activity such as increased frequency (more exercise sessions per week), duration (increase in length of exercise sessions) or intensity (increase in the exercise load such as adding hill running). Less frequently pain is reported acutely after a specific incident.

Pain behaviour

Patients with tendinopathy usually report pain localised to the tendon and is associated with tendon loading. Patellar tendinopathy is painful with jumping activities, whilst those with Achilles tendinopathy most commonly experience pain in activities related to running and hopping. In the initial stages, both Achilles and patellar tendinopathies are typically associated with pain or discomfort at the beginning of exercise that subsides with continued activity. With progression of the condition, pain is felt during exercise and can eventually lead to a cessation of activity.

One of the common features of tendinopathy, in particular Achilles tendinopathy, is the presence of morning discomfort or pain (often reported as 'stiffness' by the patient). The severity of these morning symptoms can be used to indicate the tendons' response to treatment or physical activity.

Physical assessment

In general terms, the physical assessment of both tendinopathies is similar. The most important component of the physical assessment of Achilles and patellar tendinopathies is the evaluation of muscle–tendon function, lower limb kinetic chain function and lumbo–pelvic control. We consider these to be important aspects of the assessment of any patient with lower limb tendinopathy because poor function is associated with ongoing symptoms.

A key assessment point is the evaluation of muscle–tendon function, determined by examining the strength and endurance of the muscle–tendon units that are directly and indirectly related to the pathological tendon (Table 2). With both Achilles and patellar tendinopathies, it is particularly important to assess the strength and endurance of the calf muscle–tendon unit, as it plays an important role in shock absorption of the lower limb during impact activities.⁴³ This can be done using the heel-raise test (also known as calf-raise or toe-raise test) (Table 2). For patellar tendinopathy, it is also important to assess the strength of the quadriceps and gluteal muscle groups using a decline squat test and single leg squat test respectively (Table 2).

In patients with chronic tendinopathy, quadriceps and calf muscle atrophy may be apparent and seen by observing a decrease in thigh and calf bulk.⁴² This can be measured by examining for a decrease in circumference. This is a useful clinical measurement for indicating muscle atrophy, although it must be interpreted with caution because circumference measurements are unreliable and can be influenced by both swelling and the ratio of fat versus muscle.⁴⁴

It is also important to attempt to quantify the patient's pain and level of dysfunction. The use of valid and reliable outcome measurement tools, such as the Victorian Institute of Sport Assessment (VISA) questionnaires for Achilles and patellar tendinopathies, are an important component of the initial assessment and will help to quantify the pain and functional limitations of the patient.^{45–48}

A complete lower limb biomechanical evaluation is also required in any patient with Achilles or patellar tendinopathy. In particular, assessment of abnormal lower limb alignment such as excessive foot pronation, a decreased range of motion of ankle dorsiflexion and reduced muscular flexibility³⁵ are important as they have been linked with both tendinopathies.

Finally, inspection and palpation of the tendon should be completed. Visible thickening of the tendon is common in tendon pathology especially at the midportion of the Achilles tendon. However, when tendon pathology occurs at the entheseal attachment in both the Achilles and patellar tendons, the presence of thickening may be less obvious. Both conditions are characterised by palpation tenderness, however it is important to realise that tenderness is not unusual in tendons without pathology⁴⁹ and, therefore, is an unreliable diagnostic test or outcome measure.

MONITORING IMPROVEMENT AND EVALUATING THE EFFECTIVENESS OF THE REHABILITATION PROGRAMME

The main aim of tendinopathy rehabilitation should be to decrease or abolish pain during tendon loading activities and restore normal function. An important component of any rehabilitation programme is being able to monitor progress and make appropriate adjustments to attain the best possible results. As tendinopathy is a state of failed healing, improvement in tendon morphology on radiological imaging will be slow or may not occur at all.^{19,46,50} Imaging is also not an accurate reflection of current and future symptoms because radiologically evident tendinopathy can persist for longer than symptoms.^{19,50} Similarly, tendon palpation is also ineffective in measuring progress in tendinopathy because palpation tenderness has been shown to precede symptoms and remain well after the return to full function.⁴⁹

Whilst the end goals of a rehabilitation programme (abolish pain and restore normal function) may be obvious, progress towards these goals may not be so easy to measure because there are very few clinical or radiological signs of tendon healing. As such, regular re-assessment of the patient during the rehabilitation programme is required to evaluate its effectiveness. It is recommended that reliable outcome measurement tools and specific clinical tests should be used to assess the success of the rehabilitation programme and progression towards the end goals. The following is a description of some of the outcome measurement tools available for evaluating pain and function in Achilles and patellar tendinopathies, some key clinical tests that assess the patient's functional level and the acceptable pain behaviour during the rehabilitation programme. Together, these key areas should be used to evaluate the success of the rehabilitation programme and guide the progression of exercise and function.

Outcome measures in tendinopathy

The tendinopathy rehabilitation programme needs to be constantly reviewed and modified by the treating practitioner based on the patient's pain and functional level. Standardised outcome measurement tools that are valid and reliable have an important role in determining the clinical severity of tendinopathy and monitoring progress. The VISA questionnaires for patella (original VISA) and Achilles tendons (VISA-A) have been shown to be valid and reliable outcome measurement tools.^{45–48} These outcome measurement tools evaluate pain behaviour such as morning stiffness, pain with activity and functional limitations specific to each tendinopathy. The use of these questionnaires at regular intervals during the rehabilitation programme will evaluate tendon pain and function and indicate the effectiveness of the treatment programme. However, because of the large domain of sporting function in the VISA scores, they are not sensitive to change over short periods and at least 1 month is required between each administration of the VISA score.

Functional assessment of progress in tendinopathy rehabilitation

Functional clinical tests also provide feedback on the rehabilitation programme and are more sensitive to change over shorter time periods than the VISA questionnaires. We recommend that clinicians regularly utilise such clinical tests to monitor change and evaluate the effectiveness and progress of rehabilitation. Functional tests can be used to evaluate both muscle–tendon function and pelvic/lower limb kinetic chain. Some of the key tests are described below, however the choice of test should be dependant on what the clinician is comfortable using and on the patient's pain and athletic ability.

Each assessment category should be evaluated using at least one of the recommended clinical tests or an equivalent test described in [Table 1](#). These clinical tests should measure both quality of movement and/or quantity of pain using a validated pain scale such as the Visual Analogue Scale (VAS).

The two main impairment categories that need to be assessed, as outlined in [Table 1](#), are:

1. **Muscle–tendon function:** this essentially refers to the ability of the muscle–tendon complex to generate the adequate strength and endurance required to perform functional tasks and sporting activities. This is particularly important in the rehabilitation programme as lower limb tendinopathy results in a decrease in strength of several muscles.³¹

Muscle–tendon function can be evaluated using tests that load key lower limb muscles. It is important to assess the function of the muscle–tendon units above and below the pathological tendon because any dysfunction can lead to poor load transference and increased tendon load.

The heel-raise test (also known as toe-raise or calf-raise) can be utilised to assess the strength and endurance of the calf muscle–tendon complex ([Figure 1](#)). Testing the function of the calf complex is important in both Achilles and patellar tendinopathies because of its role in attenuating forces associated with impact loading and therefore minimising the load on the pathological tendon ([Table 1](#)).

The muscle–tendon function of the quadriceps complex can be assessed using the decline squat. The decline squat ([Figure 2](#)) is a sensitive test for assessing the quadriceps function because it significantly increases patellar tendon load and produces

Table 1. Key assessment categories/tests for Achilles tendinopathy (AT) and patellar tendinopathy (PT).

Assessment category	Tendon injury	Example of clinical tests	Clinical test measurement characteristics	Description of test
Muscle—tendon function	PT	Single leg squat	1. Quality of movement — pelvic/lower limb control (as described for Hop Test) 2. Pain — VAS and knee angle where pain occurs	Patient stands on one leg and slowly lowers their body until either pain or knee flexion 90°, whilst trying to maintain an upright posture
		Decline squat (Figure 2)	1. Quality of movement — pelvic/lower limb control (as described for Hop Test) 2. Pain — VAS and knee angle where pain occurs	Patient stands on decline board angled at 25° and squats whilst trying to maintain an upright posture
	AT	Heel-raise endurance test (Figure 1)	1. Number of heel raises to fatigue or pain 2. Pain (VAS)	Standing on one foot on floor (optional; can carry a backpack with weight or on calf-raise machine). Heel-raise performed by moving from floor to full plantar flexion. Repetitive heel-raises performed to fatigue or pain
		Heel-raise strength test (Figure 1)	1. Weight carried at point of fatigue or lack of output	Standing on one foot on floor and carrying a backpack with weight or on calf-raise machine. Heel-raise performed by moving from floor to full plantar flexion. Gradually increase the weight until they are unable to do anymore because of pain or lack of power output
Function of the lower limb/pelvic kinetic chain	AT & PT	Hop test	1. Quality of hop i Pelvic/lower limb control such as excessive movement of the pelvis (e.g. inferior pelvic movement on the non-weight-bearing side, hip adduction & internal rotation of weight-bearing leg) ⁶⁷ ii Hop height compared to other leg ⁴⁴ 2. Pain — on take-off and landing using VAS	Patient stands on one leg and performs 5–6 consecutive rhythmical hops on the same spot ⁶⁷ at a hopping frequency of approximately two jumps per second. ⁴⁴
		Hop for distance	1. Distance covered with one hop 2. Quality of landing — balance and pelvic/lower limb control (as described for Hop Test)	Patient stands on one leg and performs one forward hop, trying to attain the greatest distance possible.
		Hop for height	1. Maximum height reached 2. Quality of landing — balance and pelvic/lower limb control (as described for Hop Test)	Patient stands on one leg and performs one vertical hop trying to leap as high as possible. Can be performed as either a CMJ or SJ. ⁴⁴
		Dip test Figure 3)	1. Angle of knee at loss of pelvic control such as hip adduction and lateral/anterior pelvic tilt ⁶⁷	Single leg squat with non-weight-bearing foot resting lightly on a chair positioned behind them for balance support ⁶⁷
		Single leg stand test	1. Loss of pelvic control such as inferior movement of the non-weight bearing side of pelvis ⁶⁷	Patient stands on one leg and moves the non-weight-bearing leg so it is in 90° of hip and knee flexion ⁶⁷

CMJ, Counter Movement Jump; SJ, Standing Jump.

Table 2. Rehabilitation exercises to restore functional impairments in rehabilitation stages.

Rehabilitation Stage	Functional impairment	Exercise aim	Exercise
Stage 1	Muscle–tendon function	Improve Achilles – calf strength Improve patellar tendon–quadriceps strength Improve gluteal muscle group	Heel raise exercise on a step or floor (Figures 1 and 4) Decline squats (Figure 2) Step downs Leg extension and press (machine) Lunges Leg press (machine)
Stage 2	Lower limb and pelvic kinetic chain	Promote smooth movements at the hip, knee & ankle during impact load	Hopping Skipping Jumping squats
Stage 3	Lumbo–pelvic control	Co-contraction of synergistic muscles around the pelvis & lumbar spine	Single leg bridge in supine (Figure 5) Prone 4-point bridge (Figure 6)
Stage 4	Power	Improve development for explosive movement	Skipping
Stage 5	Sport-specific exercise	Gradually increase sports-specific load to the tendon and monitor reaction to higher impact loading	Walk–jog programme Sprinting Jumping/landing Acceleration activities Change of direction activities

greater quadriceps activation (measured by electromyography) when compared to conventional single leg squats.⁵¹

These tests of muscle–tendon function provide information about the muscle–tendon units' ability to cope with load and should be used to make decisions about functional progress during the rehabilitation programme.

2. Pelvic and lower limb kinetic chain function: This is an assessment of how the pelvic/lower limb joints and muscles move in a smooth, efficient and coordinated manner, as required to perform most functional tasks and sporting activities. We recommend that kinetic chain function can be evaluated using tests such as hopping, hop for distance and hop for height. These tests have been selected as they represent practical single leg tests that load the Achilles and patellar tendons and are representative of the activities that may have contributed to the injury, such as jumping, landing or the weight-bearing component of running. The importance of some of these kinetic functions in tendinopathy, have been described by Silbernagel et al³⁴, who demonstrated that patients with Achilles tendinosis had functional deficits in hopping and hopping for height tests when compared to asymptomatic limbs.

As part of the kinetic chain function, lumbo–pelvic control must be evaluated. There are many definitions of lumbo–pelvic stability. We describe lumbo–pelvic control as the ability of the musculoskeletal system to maintain optimal alignment of the



Figure 1. Heel raise on floor.

lumbar spine and pelvis during functional weight-bearing activities. Despite the lack of scientific evidence to link a lack of lumbo–pelvic control with lower limb tendinopathy, we believe that this is an important component of the rehabilitation programme because a lack of lumbo–pelvic control has the potential to alter load distribution on the lower limb kinetic chain. We recommend the use of the two clinical tests



Figure 2. Decline Squat.



Figure 3. Dip Test.

outlined in [Table 1](#) because they can be used in patients with varying athletic ability. The dip-test ([Figure 3](#)) involves a supported single leg squat and is useful for patients with a moderate to high level of athletic ability, whilst in the single leg stand test, the patient stands balanced on one leg with the non-weight-bearing leg held at 90° hip flexion. This test is better suited to patients with lower levels of athletic ability. Assessment of lumbo–pelvic stability as observed during the decline squat and the calf-raise test will also highlight deficiencies in control.

Pain behaviour of tendinopathy

Traditionally, pain during rehabilitation was considered detrimental because it was thought to be an indication of increasing pathology, however Alfredson et al⁵² and Silbernagel et al⁵³ reported excellent results for an exercise programme that allowed pain during exercise. For these reasons, pain during exercise may not be the best guide for exercise progression.

We believe that it is important to consider the 24 hour pain behaviour when implementing and progressing rehabilitation programmes for tendinopathy. The 24 hour behaviour includes the severity and duration of pain soon after exercise and the extent of morning ‘stiffness’ or discomfort on functional tasks 24 hours later.

We recommend that it is acceptable for a slight increase in tendon discomfort or pain as a result of the rehabilitation programme, but only for a limited period of time. Ideally, no change or a decrease in pain behaviour should occur as a result of participating in the exercise programme. However, an increase in pain by 2–3 points on the Visual Analogue Scale, the day after taking part in rehabilitation exercises is acceptable, as long as it subsides within 24 hours of the exercise or activity. Any pain increase that

lasts longer than 24 hours should be viewed as a contraindication to progressing the rehabilitation programme and appropriate adjustments should be made to the rehabilitation programme to try and decrease the patient's symptoms.

THE TENDINOPATHY REHABILITATION PROGRAMME

The rehabilitation programme consists of a number of inter-related components, namely:

1. Managing tendon pain with the modification of tendon load
2. Exercise-based rehabilitation programme and adapting the tendon to increasing load
3. Additional treatment options for tendinopathy

Managing tendon pain with the modification of tendon load

Limiting tendon load is an effective method of decreasing the patient's symptoms to manageable levels and is a crucial part of the rehabilitation programme. Whilst complete cessation of all loading activities is not appropriate because it leads to weakening of the musculotendinous unit and adverse changes within the tendon itself, a partial reduction in loading will enable the patient's symptoms to decrease to a level that will allow adequate function for sport, daily activities or to perform the rehabilitation programme.⁴⁶ For example, a reduction in an athlete's training load may diminish symptoms to a level that allows the athlete to continue competing.

Once the patient's symptoms have stabilised, it is equally important to increase tendon load without increasing symptoms. This needs to be done in a gradual and organised manner by initially progressing through the different stages of the rehabilitation programme and gradually adding training and competition loads.

Exercise-based rehabilitation programme and adapting the tendon to increasing load

Exercise prescription is an important part of the rehabilitation process but has lacked high quality research to support its use. Recent evidence, however, shows that structured and specific exercise-based rehabilitation programmes for patellar and Achilles tendinopathies are successful.^{4,23,26,52–57} Whilst there are some fundamental differences between exercise protocols for Achilles and patellar tendinopathies reported in the literature, the main principles of exercise therapy are similar. In this section we will describe some of the general principles for exercise-based rehabilitation for tendinopathy and some specific exercise protocols used to treat Achilles and patellar tendinopathies.

The reader will note that there is no section on electrophysical therapeutic modalities, taping and bracing or icing as part of the treatment protocols described in this chapter. It is our belief that these treatments add very little to the rehabilitation of chronic tendinopathy. They may have a role in managing tendon pain when the athlete continues to train and play, however they will have a limited effect on long term tendon pain.

General principles for tendinopathy rehabilitation

When designing a rehabilitation exercise programme for patients with tendinopathy, it is important to consider the patients pre-injury functional/sporting level and the ultimate goal of the rehabilitation programme. The rehabilitation programme must

eventually include exercises that simulate the individual's maximal sporting function by activating the appropriate muscular activity at suitable loads, speeds and angles. This functional retraining will ensure that the tendon can manage the level of loading required to return to pre-injury levels of activity. To achieve this, the rehabilitation exercise programme must incorporate some general principles of exercise programme design, such as strength, endurance, power and a gradual return to sports-specific function. Although these exercise characteristics have been considered important components in training programmes for sporting activity, they have been largely neglected in the rehabilitation of injury. Strength, muscular endurance and power are essential components of human movement in sport and daily living activities and need to be strategically incorporated in the rehabilitation programme. Likewise sports-specific exercises that mimic loads, angles and speed of the functional and sports activity must be included in the rehabilitation programme at the appropriate stage.

Some of the important principles of a well designed rehabilitation programme are:

- There must be a regular increase in weight or resistance of exercises to progressively and systematically overload the muscle–tendon unit to maximise strength gains.
- The programme must address muscular strength–endurance by also using higher repetition and low load exercises.
- There must be an increase in the speed of exercises with the emphasis on improving muscle power.
- The programme must be maintained until full function is restored. Exercise programmes are rarely shorter than 3 months and typically need to be maintained for between 6–12 months.⁴⁶
- A maintenance programme must continue even when the athlete returns to competition.
- In a competing athlete, there may be a need for a reduction in rehabilitation loads to levels that allow the athlete to compete with minimal pain. This approach can be utilised until the end of the season and a full rehabilitation programme instituted during the off-season.⁴⁶ It is also important to consider that exercise programmes implemented whilst the athlete is training and competing are less likely to be as successful³ as those conducted when athletes are rehabilitated out of season.^{4,55}

Exercise prescription to restore functional impairments

As discussed previously, addressing dysfunctions in the muscle–tendon complex and the pelvic/lower limb kinetic chain are an important part of the tendinopathy rehabilitation programme. As there is no empirical research to guide the rehabilitation of these key areas, the following is based on our experience (Table 2). The programme outlined in Table 2 has been divided in to stages for easy application, however, these stages are overlapping and interchangeable to meet the patient's needs.

We believe that the exercise programme designed to address the identified impairments associated with lower limb tendinopathy should be implemented in a systematic order. The initial part of the exercise programme should focus on restoring muscle–tendon function of the lower limb and pelvic muscles.

Muscle–tendon unit function

Weakness and a decrease in strength/endurance in quadriceps, calf and gluteal muscle groups are common in those with chronic tendon pain and must be addressed before more complex exercises, such as kinetic chain and sports-specific exercises, are

introduced. The assessment of some of these muscle—tendon units is outlined in [Table I](#). We believe that the most effective method to rehabilitate muscle—tendon unit dysfunction is to use exercises that maximally load the target muscle—tendon group without causing adverse changes in pain behaviour.

For the quadriceps—patellar tendon complex, there is a role for open kinetic chain exercises, such as resisted leg extensions, to improve muscular strength and endurance in the early stages of rehabilitation. Closed kinetic chain exercises such as leg presses, step-downs and squats also target the quadriceps group. In particular, single-leg decline squat exercises ([Figure 2](#)) can be used as they have been shown to load the quadriceps muscle—tendon complex more than squats on a flat surface or step-down exercises.⁵¹

Improving the strength of the calf—Achilles complex is best achieved using the heel-raise exercise (see [Figures 1 and 4](#)). Shalabi et al⁵⁸ demonstrated that the heel-raise exercise, performed both concentrically and eccentrically, resulted in an immediate physiological response in both the calf muscle group and the Achilles tendon, indicating that the appropriate muscle—tendon complex is targeted effectively.

Finally, there is also a role for more traditional strength exercises, performed with gymnasium equipment, to improve muscle—tendon function. Exercises that target lower limb musculature such leg presses, hamstring curls, lunges and squatting exercises can also be used to target appropriate muscle groups around the pelvis and lower limb.

Once appropriate exercises have been selected, it is important to consider the choice of exercise characteristics, which can be varied by the clinician in an attempt to stimulate the desired neuromuscular adaptation. For instance, to improve strength the use of moderate to high resistance (load) with low repetitions is more effective, whilst high repetitions at lower resistance will help improve muscular endurance.⁵⁹

Pelvic/lower limb kinetic chain exercises

Once there is satisfactory muscle—tendon unit strength and endurance, the next stage of the rehabilitation programme is to address the more complex movements associated



Figure 4. Heel raise off a step.

with pelvic and lower limb kinetic chain function. Impairments in the lower limb and pelvic kinetic chain are usually the result of a lack of coordination and efficiency of joints and muscles involved in weight-bearing activities. It is, therefore, important that exercises prescribed involve smooth and coordinated weight-bearing activity.

Once the patient has demonstrated adequate improvement in muscle–tendon unit strength and endurance, the aim of the rehabilitation programme should be to develop muscle power. Power is a function of strength (force) and speed (velocity) of movement^{59,60} and is important in most sporting activities, particularly when a single movement sequence to produce high velocity at impact is required or in activities that require the ability to produce force quickly.⁶⁰ Power is also important in activities of daily living because most movements occur in less than 0.2 seconds, therefore qualifying as power movements.⁶⁰ Most importantly, power exercises place higher loads on the tendon portion of the muscle–tendon unit, therefore the addition of speed to a programme can increase tendon pain and loads must be carefully monitored.

Initially, simple weight-bearing exercises such as single leg squats, step downs or lunges, can be performed at a speed that is appropriate for the patients functional level. The speed of the exercise is progressively increased until the athlete becomes proficient at performing the exercise at faster speeds that are similar to athletic or functional activities. After satisfactory completion of simple faster movements, more demanding exercises such as hopping, skipping or jumping squats, can be added to improve the kinetic chain function. Skipping can be performed for short bursts with regular rest periods. For example, repeating cycles of skipping on both legs for 30 seconds and having a 60 second rest period is a good way to begin the programme. Gradually increasing the skipping time and decreasing the rest period is an ideal way to progress the exercise, as is the introduction of alternate leg and single leg skipping. Due to the increased impact forces associated with these exercises, they should be initially performed with low repetitions and sets and with 2–3 days between sessions to allow adequate recovery. As the programme progresses and the patient demonstrates an ability to perform the movement well, there can be a gradual increase in repetitions or sets. It is once again important to monitor the 24 hour pain behaviour following each session as an indication of the tendons' tolerance to such activity. It is important that the patient still maintains the strength/strength–endurance exercises performed previously in the rehabilitation programme while incorporating power exercise.

In addition to lower limb exercises, we recommend the use of low intensity exercises to target muscles traditionally involved in maintaining upright posture and spinal stiffness. Specific muscles from the abdominal, erector spinae and gluteal muscle groups are typically involved in maintaining a stable lumbar spine and pelvis during weight-bearing activity^{61–63} and should be the main focus of this part of the rehabilitation programme.

We believe that these muscle groups need to be targeted in isolation and as part of the kinetic chain. Exercises such as single leg bridging in supine (Figure 5) and four-point prone bridging exercises (Figure 6) can be used to improve lumbo–pelvic control. Exercise prescription should involve low intensity exercise that is either held isometrically for prolonged periods or performed with high^{20–30} repetitions.

Sport-specific exercise

The final part of the rehabilitation programme involves the introduction of sports-specific function. Based on good strength and power, these exercises incorporate dynamic activity to introduce load similar to the planned activity to which the patient is returning.



Figure 5. Single Leg Bridge.

Jogging is an integral part of most sporting activities and needs to be introduced at this stage of the rehabilitation programme. Initially jogging commences with a walk—jog programme that involves alternating, short periods of jogging and walking (for example 1 minute jogging followed by 1 minute walking). The walk—jog programme must commence with sessions lasting no more than 10–15 minutes in total and should be followed by 2–3 days of low tendon load activity including strength exercises to allow the tendon to adapt to the increase in load.

When the athlete can jog for more than 30 minutes without increasing their pain, more demanding sports-specific exercises including sprinting, jumping/landing, acceleration and change of direction activities are added. As these exercises involve high tendon loads, they must be introduced in a graduated and systematic manner to allow for adaptation by the muscle—tendon complex. As such, this part of the rehabilitation programme must be carefully planned in consultation with the athlete and their coach,



Figure 6. Prone Hold.

so that it involves sessions of gradually increasing intensity and volume with appropriate rest periods. As with other parts of the rehabilitation programme, careful monitoring of the patient's pain behaviour over a 24 hour period must take place to evaluate the tendons' ability to cope with such activity.

Eccentric exercise and tendinopathy

Curwin & Stanish⁶⁴ first proposed that eccentric exercise was an effective method of treating tendinopathy and since then there has been an increasing body of evidence to support the use of this approach. Eccentric based exercise programmes have been shown to be the most effective way of treating lower limb tendinopathy related pain and dysfunction^{4,17,23,26,52–57} and are the cornerstone to exercise rehabilitation for Achilles and patellar tendinopathies. In particular, the use of eccentric heel drops for Achilles tendinopathy (Figures 1 and 4) and eccentric single leg squats on a step or decline board (Figure 2), for patellar tendinopathy, have known efficacy.

Whilst the use of eccentric exercise to treat lower limb tendinopathies is not disputed, there is still some debate regarding the precise exercise protocol that is required for optimal results. In particular there are some differences reported in exercise characteristics amongst some of the key eccentric tendinopathy programmes. For example the Curwin & Stanish⁶⁴ programme found that using graduated increases of moderate eccentric loads and changes in exercise speed, resulted in symptomatic improvement in patients with tendon pathology over a 6-week period. Alfredson et al⁵² also used eccentric heel drops on a step and focused on increasing heavy load (weight). Using this approach, they demonstrated significant improvements in pain and function in recreational athletes with chronic Achilles tendinopathy. They reported that 100% of their cohort of recreational athletes, who were unable to run due to Achilles tendon pain prior to the programme, were able to return to running with a significant decrease in pain after their 12 week programme. The Alfredson et al⁵² approach differed to that of the Curwin & Stanish⁶⁴ protocol in that it utilised heavier resistance, did not incorporate speed into the exercise programme and allowed pain during exercise.

Young et al⁴ and Purdam et al⁵⁵ reported that performing eccentric single leg squats on a 25° decline board was more effective than performing single leg exercises on a flat surface. It is possible that the superior results using the decline board were a result of the increased load on the patellar tendon caused by the decline angle, when compared with traditional flat squats or step down exercises.⁵¹

These differences in exercise protocols indicate that the mechanisms that underpin good results with eccentric programmes is unclear. Despite this, we believe that clinicians should incorporate an eccentric component into their exercise prescription, either a pre-tested 'recipe' programme or as part of an individualised programme. The programme proposed in this chapter incorporates eccentric movement as part of every exercise but does not use specific eccentric exercises.

Limitation of eccentric exercise

The timing of an eccentric exercise programme in athletes may be vital. Young et al⁴ reported good outcomes using an eccentric exercise programme to treat high level volleyball players. In contrast however, Visnes et al³ found no difference between an eccentric exercise programme and no intervention in a group of high level volleyball players. Whilst these two studies used a similar exercise protocol involving eccentric decline squats, there was one major difference between them that may explain the

disparity of the results. One study⁴ was conducted out of the competitive season (pre-season), whilst the other³ was conducted during the competitive season when players were still expected to train and compete. These results would suggest that the eccentric exercise may be less effective when implemented during a competitive season in high level athletes.

Another potential limitation of the eccentric exercise protocol may be in the management of chronic insertional Achilles tendinopathy. Fahlstrom et al⁶⁵ reported good clinical results using an eccentric calf muscle exercise programme in patients with chronic mid-portion Achilles tendinosis, but found that the same programme was less effective for treating those patients with insertional tendinopathy. Therefore, the use of eccentric exercise as the sole treatment for insertional Achilles tendinopathy needs to be viewed with caution. This limitation of eccentric exercise with insertional tendinopathy does appear to be limited to the Achilles tendon, since eccentric treatment of other insertional tendinopathies at the patellar^{4,55} and adductor tendons⁶⁶ have demonstrated good clinical results. Clinically we believe that performing eccentric heel-raise exercises on a flat surface (rather than over a step) may be more beneficial in treating recalcitrant insertional Achilles tendinopathy.

Effect of eccentric exercise on tendon pathology

Eccentric exercise is beneficial in reducing Achilles and patellar tendon pain; however the effect it has on pathological tendons is still unclear. It has been proposed that eccentric exercise may better equip the muscle–tendon unit to cope with the forces associated with physical activity or that eccentric exercise may directly improve tendon pathology.⁶ Recent research has demonstrated that eccentric exercise may have a positive affect on tendon thickness and structure^{17,23} in patients with Achilles tendinopathy. Ohberg et al²³ found that a group of physically active subjects with symptomatic thickened Achilles tendons had a significant decrease in tendon thickness and improvement in tendon structure (measured by ultrasonography) at follow-up, 3.8 years after performing an eccentric exercise programme. They also reported that 88% of patients who took part in the eccentric exercise programme were satisfied with their treatment and were physically active at follow-up, indicating a positive association between the good clinical results and the decrease in tendon structure and thickness. By contrast, subjects that had residual tendon pain at follow-up were associated with unchanged tendon structure.

Another benefit of eccentric exercise in pathological tendons is a decrease in neovascularisation. As reported earlier, the presence of neovessels is correlated with painful Achilles and patellar tendinopathies.¹⁸ Ohberg et al¹⁷ reported that there was neovascularisation in the tendons of all their subjects with painful Achilles tendinopathy before commencing an eccentric exercise treatment. After 12 weeks of eccentric exercise, they reported excellent clinical results, with 88% of patients reporting a good clinical outcome 28 months after treatment had ceased. Of those patients who had good clinical outcomes, 89% had no neovascularisation at 28 months post-treatment. This suggests that eccentric exercise may decrease neovascularisation.

Additional treatment options for tendinopathy

Factors identified in the assessment need to be addressed during the rehabilitation programme. Some of the common factors associated with Achilles and patellar tendinopathies are:

1. Excessive foot pronation or supination during weight-bearing activities is thought to adversely affect lower limb load absorbing function and potentially increase tendon loading. Once identified the patient should be referred to a podiatrist for a more detailed assessment and possible orthotic intervention.
2. Any decrease in ankle dorsiflexion needs to be identified and treated. To maximise the benefit of ankle mobilisation, it is important to identify the most likely cause of the decrease in range. For example, if calf muscle length is restricting dorsi-flexion, then treatment could involve muscle stretching or night splints to improve range by altering the viscoelastic properties. Conversely, if the source of restriction is joint related, treatment may involve joint mobilisation.
3. Finally, a decreased range of hamstring and quadriceps flexibility has been associated with patellar tendinopathy and should be managed with regular stretching. As with other muscle stretches, it is important to educate the patient that the expected time frame for significant changes in muscle length can be many weeks or months.

CONCLUSION

We have described the pathophysiology of chronic Achilles and patellar tendinopathy as a state of failed healing. This healing response and the unclear source of tendon pain means that tendinopathy rehabilitation is often difficult and prolonged. The initial part of the rehabilitation process should include a thorough clinical assessment to establish the functional status of the patient's lower limb muscle—tendon function, lumbo—pelvic control, lower limb kinetic chain function and to identify other factors that may contribute to symptoms.

The tendinopathy rehabilitation programme is based around an exercise programme that aims to address deficiencies in muscle—tendon function and pelvic/lower limb kinetic chain. Rehabilitation must also include exercises to develop muscle power and sports-specific function.

The nature of tendon healing means that the rehabilitation programme needs to be performed for many months and should be maintained even when the patient has returned to their desired functional and sporting activity.

Practice points

- Chronic tendon pathology is considered to be a state of failed healing
- The presence of neurovascular bundles in pathological tendons is thought to be related to tendon pain
- Assessment of patients with lower limb tendinopathy must involve assessment of muscular strength and endurance, as well as pelvic/lower limb kinetic chain function
- Management of chronic tendinopathy is exercise based. Exercise with an eccentric component should be incorporated where possible
- The rehabilitation programme should be aimed at restoring strength, kinetic chain function, power and sports-specific function
- The rehabilitation programme should continue even when the person returns to full function

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